

- 23 Find the coordinates of the turning point on the graph of $y = 7 - 2x - x^2$.

$$= -x^2 - 2x + 7$$

$$= -\frac{1}{2}(x+1)^2 + 8$$

$$= -x^2 - 2x + \frac{1}{2} + \frac{1}{2} + 7$$

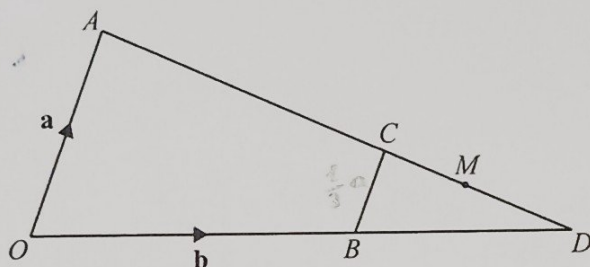
$$= -\frac{1}{2}(x^2 + 2x - 7)$$

$$= -\frac{1}{2}[x^2 + 2x + 1 - 7 - 1]$$

$$= -\frac{1}{2}[(x+1)^2 - 8]$$

$$(\dots\dots\dots -1 \dots\dots\dots, \dots\dots\dots 8 \dots\dots\dots) [4]$$

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NOT TO
SCALE

In the diagram, OBD and ACD are straight lines.

O is the origin, the position vector of A is $\mathbf{a}$ and the position vector of B is $\mathbf{b}$.

$$\overrightarrow{BC} = \frac{1}{3} \overrightarrow{OA}$$

M is the midpoint of CD .

Find the position vector of M .

Give your answer in terms of $\mathbf{a}$ and $\mathbf{b}$, in its simplest form.

$$\begin{aligned} & \overrightarrow{AM} \\ &= \overrightarrow{AO} + \overrightarrow{OB} + \overrightarrow{BC} \\ &= -\mathbf{a} + \mathbf{b} + \frac{1}{3}\mathbf{a} \\ &= \frac{-2}{3}\mathbf{a} + \mathbf{b} \end{aligned}$$

..... [4]

Questions 25 and 26 are printed on the next page.